LECTURE: MULTIPLE LINEAR REGRESSION

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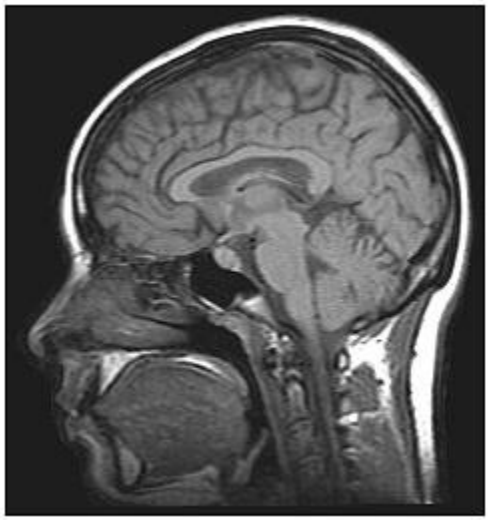
Example 1: IQ and Physical Characteristics

Are a person's brain size and body size predictive of his or her intelligence?

Interested in answering the above research question, some researchers (Willerman,*et al*, 1991) collected the following data (iqsize.txt) on a sample of n = 38 college students:

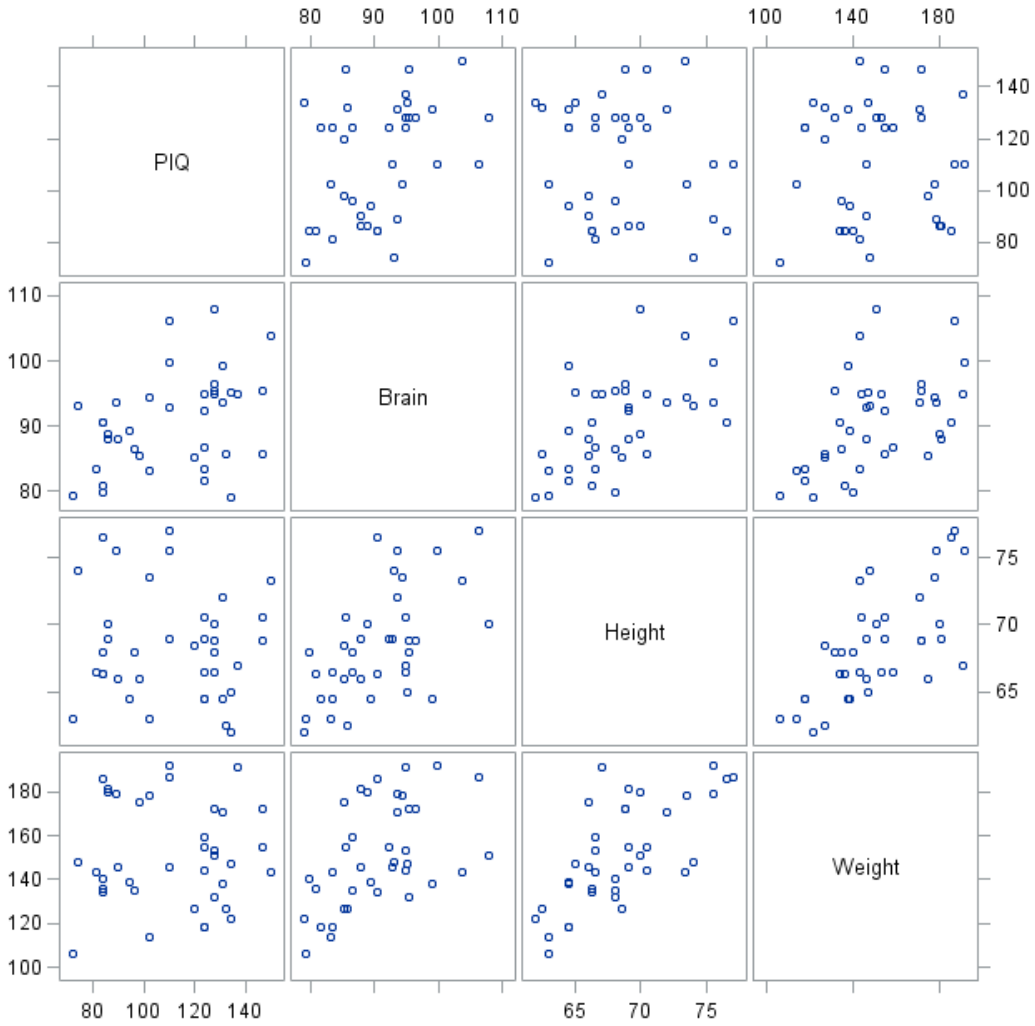
- Response (y): Performance IQ scores (**PIQ**) from the revised Wechsler Adult Intelligence Scale. This variable served as the investigator's measure of the individual's intelligence.
- Potential predictor (x_1): Brain size based on the count obtained from **MRI** scans (given as count/10,000).
- Potential predictor (x_2): **Height** in inches.
- Potential predictor (x_3): **Weight** in pounds.



Scatter plot matrix

A common way of investigating the relationships among all of the variables is by way of a "**scatter plot matrix**.“ Now, what does a scatter plot matrix tell us? Of course, one use of the plots is simple data checking. Are there any egregiously erroneous data errors? The scatter plots also illustrate the "**marginal relationships**" between each pair of variables *without regard to the other variables*. For example, it appears that brain size is the best single predictor of PIQ, but none of the relationships are particularly strong. In multiple linear regression, the challenge is to see how the response *y* relates to all three predictors *simultaneously*.

Scatter plot matrix



The Model

We always start a regression analysis by formulating a model for our data. One possible multiple linear regression **model with three quantitative predictors** for our brain and body size example is:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \varepsilon_i$$

where:

- y_i is the intelligence (**PIQ**) of student i
- x_{i1} is brain size (**MRI**) of student i
- x_{i2} is the height (**Height**) of student i
- x_{i3} is the weight (**Weight**) of student i

and the **independent** error terms ε_i follow a **normal** distribution with mean 0 and **equal variance** σ^2.

Research questions

- Which, if any, predictors — brain size, height, or weight — explain some of the variation in intelligence scores? (Conduct hypothesis tests for individually testing whether each slope parameter could be 0.)
- What is the effect of brain size on PIQ, after taking into account height and weight? (Calculate and interpret a confidence interval for the brain size slope parameter.)
- What is the PIQ of an individual with a given brain size, height, and weight? (Calculate and interpret a prediction interval for the response.)

The Multiple Linear Regression Model

- A population model for a multiple regression model that relates a *y*-variable to *p* -1 predictor variables is written as

$$y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \cdots + \beta_{p-1} x_{i,p-1} + \varepsilon_i$$

- We assume that the ε_i have a normal distribution with mean 0 and constant variance σ^2. These are the same assumptions that we used in simple regression with one *x*-variable.
- The subscript i refers to the ith individual or unit in the population. In the notation for the *x*-variables, the subscript following i simply denotes which *x*-variable it is.

Estimates of the Model Parameters

- The estimates of the β coefficients are the values that minimize the sum of squared errors for the sample. The exact formula for this is given in the next section on matrix notation.
- The letter b is used to represent a sample estimate of a β coefficient. Thus b_0 is the sample estimate of β_0, b_1 is the sample estimate of β_1, and so on.
- $MSE = \frac{SSE}{n-p}$ estimates σ^2, the variance of the errors. In the formula, $\boldsymbol{n}$ **= sample size**, $\boldsymbol{p}$ **= number of** $\boldsymbol{\beta}$ **coefficients** in the model (including the intercept) and $\boldsymbol{SSE} =$ **sum of squared errors**. Notice that for simple linear regression $p = 2$. Thus, we get the formula for MSE that we introduced in that context of one predictor.
- In the case of two predictors, the estimated regression equation yields a plane (as opposed to a line in the simple linear regression setting). For more than two predictors, the estimated regression equation yields a hyperplane.

Interpretation of the Model Parameters

- Each β coefficient represents the change in the mean response, $E(y)$, per unit increase in the associated predictor variable when all the other predictors are held constant.
- For example, β_1 represents the change in the mean response, $E(y)$, per unit increase in x_1 when x_2, x_3, ..., x_{p-1} are held constant.
- The intercept term, β_0, represents the mean response, $E(y)$, when all the predictors x_1, x_2, x_3, ..., x_{p-1}, are all zero (which may or may not have any practical meaning).
- A **predicted value** is calculated as

$$\hat{y}_i = b_0 + b_1 x_{i,1} + b_2 x_{i,2} + \cdots b_{p-1} x_{i,p-1}$$

where the b values come from statistical software and the x-values are specified by us.

- A **residual** (**error**) term is calculated as $e_i = y_i - \hat{y}_i$, the difference between an actual and a predicted value of y.
- A **plot of residuals versus predicted values** ideally should resemble a horizontal random band. Departures from this form indicates difficulties with the model and/or data.

ANOVA Table

Source	df	SS	MS	F
Regression	$p - 1$	SSR	MSR = SSR / $(p - 1)$	MSR / MSE
Error	$n - p$	SSE	MSE = SSE / $(n - p)$	
Total	$n - 1$	SSTO		

Coefficient of Multiple Determination

- As in simple linear egression,

$$R^2 = \frac{SSR}{SSTO} = 1 - \frac{SSE}{SSTO}$$

- Represents the proportion of variation in y (about its mean) "explained" by the multiple linear regression model with predictors, $x_1, x_2, \ x_3, \ldots, \ x_{p-1}$.
- Additional variables will make R^2 go up; so cannot really use R^2 to determine whether a variable should be added.

Adjusted R^2

$$R_a^2 = 1 - \frac{MSE}{MSTO} = 1 - \left(\frac{n-1}{n-p}\right)\frac{SSE}{SSTO}$$

- Recall mean squares are SS adjusted by degrees of freedom
- R_a^2 can increase or decrease when a new variable is introduced into the model; depending on whether the decrease in SSE is offset by the lost degree of freedom.
- R_a^2 can be used to decide if variables are important in a model.

Example 1: IQ and Physical Characteristics

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	111.35361	62.97110	1.77	0.0860
Brain	1	2.06037	0.56345	3.66	0.0009
Height	1	-2.73193	1.22943	-2.22	0.0330
Weight	1	0.00055994	0.19707	0.00	0.9977

Root MSE	19.79439	R-Square	0.2949
Dependent Mean	111.34211	Adj R-Sq	0.2327
Coeff Var	17.77799		

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	5572.74444	1857.58148	4.74	0.0072
Error	34	13322	391.81789		
Corrected Total	37	18895			

Example 1: IQ and Physical Characteristics

```
proc print data=iq;

proc sgscatter data=iq;
matrix PIQ Brain Height Weight;

proc reg data=iq;
model PIQ=Brain Height Weight;
run;
```

Matrix Formulation

In the multiple regression setting, because of the potentially large number of predictors, it is more efficient to use matrices to define the regression model and the subsequent analyses. Consider the following simple linear regression function:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i \qquad \text{for } i = 1, \ldots, n$$

If we actually let $i = 1, \ldots, n$, we see that we obtain n equations:

$$\begin{aligned} y_1 &= \beta_0 + \beta_1 x_1 + \epsilon_1 \\ y_2 &= \beta_0 + \beta_1 x_2 + \epsilon_2 \\ &\vdots \\ y_n &= \beta_0 + \beta_1 x_n + \epsilon_n \end{aligned}$$

we can instead formulate the above simple linear regression function in matrix notation:

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

$$Y = X\beta + \varepsilon$$

The Multiple Regression Model

Assuming that the model is

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k + \varepsilon,$$

the n-tuples of observations are also assumed to follow the same model. Thus they satisfy

$$\begin{aligned}
y_1 &= \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \ldots + \beta_k x_{1k} + \varepsilon_1 \\
y_2 &= \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \ldots + \beta_k x_{2k} + \varepsilon_2 \\
&\vdots \qquad\qquad\qquad\qquad\qquad\qquad \vdots \\
y_n &= \beta_0 + \beta_1 x_{n1} + \beta_2 x_{n2} + \ldots + \beta_k x_{nk} + \varepsilon_n.
\end{aligned}$$

These n equations can be written as

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_{11} & x_{12} \cdots x_{1k} \\ 1 & x_{21} & x_{22} \cdots x_{2k} \\ \vdots & \vdots & \vdots \quad \vdots \\ 1 & x_{n1} & x_{n2} \cdots x_{nk} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

or $y = X\beta + \varepsilon.$

Assumptions in multiple linear regression model

(i) $E(\varepsilon) = 0$

(ii) $E(\varepsilon\varepsilon') = \sigma^2 I_n$

(iii) $Rank(X) = k$

(iv) X is a non-stochastic matrix

(v) $\varepsilon \sim N(0, \sigma^2 I_n)$.

Least Squares Estimate

* For given β the residuals can be written as

$$e_i = y_i - \boldsymbol{x}_i\boldsymbol{\beta}$$

where $\boldsymbol{x}_i$ is the ith row of $\boldsymbol{X}$.

* Then we can write

$$S(\boldsymbol{\beta}) = \sum_{i=1}^{n}(y_i - \boldsymbol{x}_i\boldsymbol{\beta})^2 = (\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta})^t(\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta})$$

* On taking derivatives and setting them equal to 0 we get the least squares estimates

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^t\boldsymbol{X})^{-1}\boldsymbol{X}^t\boldsymbol{Y}$$

Estimation of σ^2

* The fitted values from the regression are $\hat{\boldsymbol{Y}} = \boldsymbol{X}\hat{\beta}$.

* The residuals are

$$\boldsymbol{e} = \boldsymbol{Y} - \hat{\boldsymbol{Y}} = \boldsymbol{Y} - \boldsymbol{X}\hat{\beta}$$

* As in the simple model, we can find an unbiased estimator of σ^2 by looking at the sum of squared residuals

$$\text{SSE} = \boldsymbol{e}^t\boldsymbol{e} = (\boldsymbol{Y} - \boldsymbol{X}\hat{\beta})^t(\boldsymbol{Y} - \boldsymbol{X}\hat{\beta}).$$

* The estimator is

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n-(p+1)}.$$

Properties of the Estimators

1. $E(\hat{\beta}|X) = \beta$
2. $Var(\hat{\beta}|X) = \sigma^2(X'X)^{-1}$
3. $E(\hat{\sigma}^2|X) = \sigma^2$
4. It is the best linear unbiased estimator (BLUE) of β

Standard Errors and Sampling Distributions

* The estimated Variance-Covariance matrix for β is given by

$$\widehat{\mathrm{Var}}(\hat{\beta}) \;=\; \hat{\sigma}^2(\boldsymbol{X}^t\boldsymbol{X})^{-1} \;=\; \sigma^2\boldsymbol{C}.$$

* The standard errors can be found by looking at the diagonal elements of this matrix.

$$\mathrm{se}(\hat{\beta}_j) \;=\; \hat{\sigma}\sqrt{c_{jj}}$$

* These standard errors are always output as part of the SAS output from a regression.

* The Sampling Distributions are

$$\frac{\hat{\beta}_j - \beta_j}{\mathrm{se}(\hat{\beta}_j)} \;\sim\; t_{n-(p+1)} \qquad j = 0, 1, \ldots, p$$

Individual Confidence Intervals

* Confidence intervals for individual parameters are given by

$$\widehat{\beta}_j \pm t_{(n-p-1;\alpha/2)}\mathrm{se}(\widehat{\beta}_j) \qquad j = 0, \ldots, p$$

* As with the estimates, these are to be interpreted as confidence intervals for the additional effect of X_j when all of the other covariates are included in the model.

Individual Hypothesis Tests

* Similarly we can test if $\beta_j = \beta_j^0$ using the test statistic

$$t_j = \frac{\beta_j - \beta_j^0}{\text{se}(\hat{\beta}_j)}.$$

* The p-value for this test can be found by comparing the test statistic to the t_{n-p-1} distribution.

* It is important to remember that this is a test of the additional contribution of X_j when all of the other covariates are in the model.

Example 2: Hospital Data

Data from $n = 113$ hospitals in the United States are used to assess factors related to the likelihood that a hospital patients acquires an infection while hospitalized. The variables here are y = infection risk, x_1 = average length of patient stay, x_2 = average patient age, x_3 = measure of how many x-rays are given in the *hospital.txt*.

Our regression model is

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \varepsilon_i$$

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	1.00116	1.31472	0.76	0.4480
Stay	1	0.30818	0.05940	5.19	<.0001
Age	1	-0.02301	0.02352	-0.98	0.3301
Xray	1	0.01966	0.00576	3.41	0.0009

SAS code

```
proc print data=hs;


proc reg data=hs;
model InfctRsk=Stay Age Xray;
run;
```

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SAS code

```
proc import datafile="\\filer01\wzhou2\My Documents\courses\8110\data\hospital.txt" out=hs;
getnames=yes;

proc print data=hs;


proc reg data=hs;
model InfctRsk=Stay Age Xray;
run;
```

Three Types of Hypotheses

Given a multiple linear regression model,

$$y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \cdots + \beta_{p-1} x_{i,p-1} + \varepsilon_i$$

We want to consider the following three tests:

- Overall test
- Test for a single variable
- Test for a group of variables

Overall test

- H_0: $\beta_1 = \beta_2 = \cdots = \beta_{p-1} = 0$
- H_a: $\beta_j \neq 0 \; for \; at \; least \; one \; j, j = 1, \cdots, p-1$

- Rejection of H_0 implies that at least one of the repressors, $x_1, x_2, \cdots, x_{p-1}$ contributes significantly to the model.

- We will use a generalization of the F-test in simple linear regression to test this hypothesis.

Significance Testing of Each Variable

Within a multiple regression model, we may want to know whether a particular *x*-variable is making a useful contribution to the model. That is, given the presence of the other *x*-variables in the model, does a particular *x*-variable help us predict or explain the *y*-variable? For instance, suppose that we have three *x*-variables in the model. The general structure of the model could be

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \varepsilon_i$$

As an example, to determine whether variable x_1 is a useful predictor variable in this model, we could test

$$H_0 : \beta_1 = 0; \qquad H_a : \beta_1 \neq 0.$$

If the null hypothesis above were the case, then a change in the value of x_1 would not change y, so y and x_1 are not linearly related. Also, we would still be left with variables x_2 and x_3 being present in the model. When we cannot reject the null hypothesis above, we should say that we do not need variable x_1 in the model given that variables x_2 and x_3 will remain in the model. In general, the interpretation of a slope in multiple regression can be tricky. Correlations among the predictors can change the slope values dramatically from what they would be in separate simple regressions.

Significance Testing of Each Variable

To carry out the test, statistical software will report *p*-values for all coefficients in the model. Each *p*-value will be based on a *t*-statistic calculated as

t^*= (sample coefficient - hypothesized value) / standard error of coefficient.

For our example above, the *t*-statistic is:

$$t^* = \frac{b_1 - 0}{se(b_1)} = \frac{b_1}{se(b_1)}$$

This is a partial test because b_1 depends on all of the other predictors that are in the model. Thus, this is a test of the contribution of x_1 given the other predictors in the model.

Example

Suppose we want to test

$$H_0 : \beta_1 = 0; \qquad H_a : \beta_1 \neq 0,$$

given x_2, x_3 in the model.

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	1.00116	1.31472	0.76	0.4480
Stay	1	0.30818	0.05940	5.19	<.0001
Age	1	-0.02301	0.02352	-0.98	0.3301
Xray	1	0.01966	0.00576	3.41	0.0009

$$t = \frac{0.308}{0.059} = 5.19, \qquad P - value < .0001$$

We reject the null hypothesis and conclude that Stay is significant given Age and Xray in the model.

Test for a group of variables

Does the addition of some *group* of independent variables of interest add significantly to the prediction of y obtained through other independent variables already in the model?

$$y_i = \beta_0 + x_{i1}\beta_1 + \cdots + x_{i,p-1}\beta_{p-1} + x_{ip}\beta_p + \epsilon_i$$

Test for a group of variables

Consider the researcher's question: "Is the infection risk significantly (linearly) related to the patient age and the amount of x-ray after controlling for the length of stay?" Are you convinced that testing the following hypotheses helps answer the research question?

- $H_0 : \beta_2 = \beta_3 = 0$
- H_A : At least one $\beta_i \neq 0$ (for i = 2, 3)

In this case, the researchers are interested in testing whether a *subset* (more than one, but not all) of the slope parameters are simultaneously zero. We will learn a general linear F-test for testing such a hypothesis.

The General Linear Test

The "**general linear test**" involves three basic steps, namely:

1. Define a larger **full model**. (By "larger," we mean one with more parameters.)
2. Define a smaller **reduced model**. (By "smaller," we mean one with fewer parameters.)
3. Use an ***F*-statistic** to decide whether or not to reject the smaller reduced model in favor of the larger full model.

As you can see by the wording of the third step, the null hypothesis always pertains to the reduced model, while the alternative hypothesis always pertains to the full model.

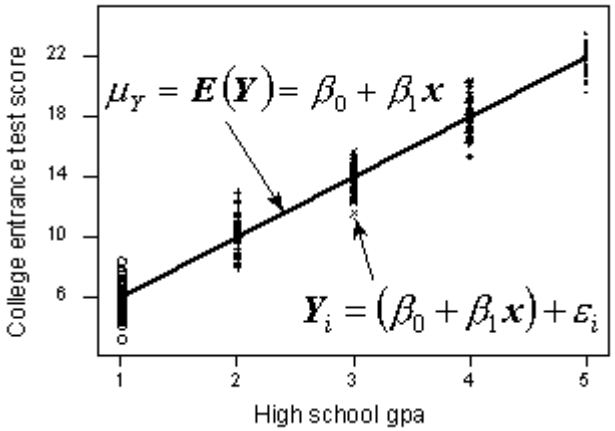
The easiest way to learn about the general linear test is to first go back to what we know, namely the simple linear regression model.

The full model

The "**full model**", which is also sometimes referred to as the "**unrestricted model**," is the model thought to be most appropriate for the data. For simple linear regression, the full model is:

$$y_i = (\beta_0 + \beta_1 x_{i1}) + \epsilon_i$$

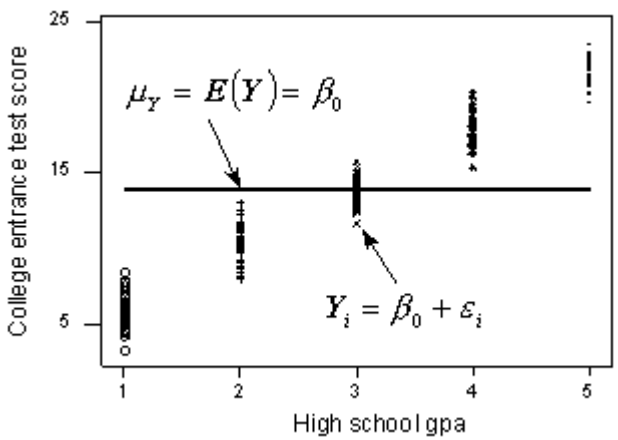
Here's a plot of a hypothesized full model for a set of data:



The reduced model

The "**reduced model**," which is sometimes also referred to as the "**restricted model**," is the model described by the null hypothesis H_0. For simple linear regression, a common null hypothesis is $H_0 : \beta_1 = 0$. In this case, the reduced model is obtained by "zeroing-out" the slope β_1 that appears in the full model. That is, the reduced model is:

$$y_i = \beta_0 + \epsilon_i$$



The test

How do we decide if the reduced model or the full model does a better job of describing the trend in the data when it can't be determined by simply looking at a plot? What we need to do is to quantify how much error remains after fitting each of the two models to our data. That is, we take the general linear test approach:

- "**Fit the full model**" to the data.
  - Obtain the least squares estimates of β_0 and β_1.
  - Determine the error sum of squares, which we denote "$SSE(F)$."
- "**Fit the reduced model**" to the data.
  - Obtain the least squares estimate of β_0.
  - Determine the error sum of squares, which we denote "$SSE(R)$."

The test

Where are we going with this general linear test approach? In short:

- The general linear test involves a comparison between $SSE(R)$ and $SSE(F)$.
- $SSE(R)$ can never be smaller than $SSE(F)$. It is always larger than (or possibly the same as) $SSE(F)$.
  - If $SSE(F)$ is close to $SSE(R)$, then the variation around the estimated full model regression function is almost as large as the variation around the estimated reduced model regression function. If that's the case, it makes sense to use the simpler reduced model.
  - On the other hand, if $SSE(F)$ and $SSE(R)$ differ greatly, then the additional parameter(s) in the full model substantially reduce the variation around the estimated regression function. In this case, it makes sense to go with the larger full model.

The Test Statistic

$$F^* = \left(\frac{SSE(R) - SSE(F)}{df_R - df_F}\right) \div \left(\frac{SSE(F)}{df_F}\right)$$

The F-statistic intuitively makes sense — it is a function of $SSE(R)$-$SSE(F)$, the difference in the error between the two models. The degrees of freedom — denoted df_R and df_F — are those associated with the reduced and full model error sum of squares, respectively.

We use the general linear F-statistic to decide whether or not:

- to reject the null hypothesis H_0: The reduced model
- in favor of the alternative hypothesis H_A: The full model

In general, we reject H_0 if F^* is large — or equivalently if its associated P-value is small.

The test applied to the simple linear regression model

The full model is:

$$y_i = (\beta_0 + \beta_1 x_{i1}) + \epsilon_i$$

and the reduced model is:

$$y_i = \beta_0 + \epsilon_i$$

Therefore, the appropriate null and alternative hypotheses are specified either as:

- H_0: $y_i = \beta_0 + \varepsilon_i$
- H_A: $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$

or as:

- H_0: $\beta_1 = 0$
- H_A: $\beta_1 \neq 0$

The test applied to the simple linear regression model

$F^* = \left(\frac{SSE(R)-SSE(F)}{df_R-df_F}\right) \div \left(\frac{SSE(F)}{df_F}\right)$	
⬇	
$df_R = n - 1$ $df_F = n - 2$	$SSE(R) = SSTO$ $SSE(F) = SSE$
⬇	
$F^* = \left(\frac{SSTO-SSE}{(n-1)-(n-2)}\right) \div \left(\frac{SSE}{(n-2)}\right) = \frac{MSR}{MSE}$	

Testing all slope parameters equal 0

To answer the research question: "Is the regression model containing at least one predictor useful in predicting the response?," we test the hypotheses:

- $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$
- H_A : At least one $\beta_i \neq 0$ (for i = 1, 2, 3)

In this case, the **full model** is:

$$y_i = (\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3}) + \epsilon_i$$

The error sum of squares for the full model, *SSE*(*F*), is just the usual error sum of squares, *SSE*, that appears in the analysis of variance table. Because there are 4 parameters in the full model, the number of error degrees of freedom associated with the full model is $df_F = n - 4$.

Testing all slope parameters equal 0

The **reduced model** is

$$y_i = \beta_0 + \epsilon_i$$

Therefore, the error sum of squares for the reduced model, *SSE*(*R*), is just the total sum of squares, *SSTO*, that appears in the analysis of variance table. Because there is only one parameter in the reduced model, the number of error degrees of freedom associated with the reduced model is $df_R = n - 1$.

The test. Upon plugging in the above quantities, the general linear *F*-statistic:

$$F^* = \frac{SSE(R) - SSE(F)}{df_R - df_F} \div \frac{SSE(F)}{df_F}$$

becomes the usual "**overall *F*-test**":

$$F^* = \frac{SSR}{3} \div \frac{SSE}{n-4} = \frac{MSR}{MSE}$$

Overall F-test

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	73.09897	24.36632	20.70	<.0001
Error	109	128.28085	1.17689		
Corrected Total	112	201.37982			

$$F = \frac{24.366}{1.177} = 20.70$$

- There is sufficient evidence ($F = 20.70$, $P < 0.001$) to conclude that at least one of the slope parameters is not equal to 0.
- In general, to test that *all* of the slope parameters in a multiple linear regression model are 0, we use the overall F-test reported in the analysis of variance table.

Partial F-Test

Consider the following hypotheses

- $H_0 : \beta_2 = \beta_3 = 0$
- H_A : At least one $\beta_i \neq 0$ (for i = 2, 3)

The full model

$$y_i = (\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3}) + \epsilon_i$$

The reduced model

$$y_i = (\beta_0 + \beta_1 x_{i1}) + \epsilon_i$$

Partial F-Test

The test. The general linear statistic:

$$F^* = \frac{SSE(R) - SSE(F)}{df_R - df_F} \div \frac{SSE(F)}{df_F}$$

becomes a partial F-test:

$$F^* = \frac{SSR(x_2, x_3|x_1)}{2} \div \frac{SSE(x_1, x_2, x_3)}{n-4} = \frac{MSR(x_2, x_3|x_1)}{MSE(x_1, x_2, x_3)}$$

Partial F-Test

Parameter Estimates						
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Type I SS
Intercept	1	1.00116	1.31472	0.76	0.4480	2143.03018
Stay	1	0.30818	0.05940	5.19	<.0001	57.30511
Age	1	-0.02301	0.02352	-0.98	0.3301	2.07506
Xray	1	0.01966	0.00576	3.41	0.0009	13.71880

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	73.09897	24.36632	20.70	<.0001
Error	109	128.28085	1.17689		
Corrected Total	112	201.37982			

$$SSR(x_2, x_3|x_1) = 2.08 + 13.72 = 15.80,\ MSE = 1.18,$$

$$F = \frac{15.80/2}{1.18} = 6.70,\ \text{P-value} = 0.0018.$$

With 112 data points, we likely have the power required to conclude that Age and Xray are useful in the model that already contains Stay.

Partial F-Test Using SAS

Can obtain this test in SAS using a test statement:

```
proc reg data=hs;
model InfctRsk= Stay Age Xray;
TEST1: test Age=0, Xray=0;
run;
```

Test TEST1 Results for Dependent Variable InfctRsk				
Source	DF	Mean Square	F Value	Pr > F
Numerator	2	7.89693	6.71	0.0018
Denominator	109	1.17689		

Confidence intervals in multiple linear regression

- Confidence interval for mean response
- Prediction interval

Confidence interval for mean response

In this section, we are concerned with the confidence interval, called a "***t*-interval**," for the mean response μ_Y when the predictor values are $\mathbf{X}_h = (1, X_{h,1}, X_{h,2}, \ldots, X_{h,p-1})^{\mathrm{T}}$. The general formula in words is as always:

Sample estimate ± (*t*-multiplier × standard error)

First we define the standard error of the fit at $\mathbf{X}_h$ given by:

$$\mathrm{se}(\hat{y}_h) = \sqrt{\mathrm{MSE}(\mathbf{X}_h^{\mathrm{T}}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}_h)}.$$

and the confidence interval is:

$$\hat{y}_h \pm t_{(\alpha/2, n-p)} \times \mathrm{se}(\hat{y}_h)$$

where:

- $\hat{y}_h$ is the "**fitted value**" or "**predicted value**" of the response when the predictor values are $\mathbf{X}_h$.
- $t_{(\alpha/2, n-p)}$ is the "***t*-multiplier**." Note that the *t*-multiplier has *n*-p degrees of freedom because the confidence interval uses the mean square error (*MSE*) whose denominator is *n*-*p*.

Factors affecting the width of the t-interval for the mean response μ_Y

As always, the formula is useful for investigating what factors affect the width of the confidence interval for μ_Y.

- **As the mean square error (*MSE*) decreases, the width of the interval decreases.** Since *MSE* is an estimate of how much the data vary naturally around the unknown population regression hyperplane, we have little control over *MSE* other than making sure that we make our measurements as carefully as possible.
- **As we decrease the confidence level, the *t*-multiplier decreases, and hence the width of the interval decreases.** In practice, we wouldn't want to set the confidence level below 90%.
- **As we increase the sample size *n*, the width of the interval decreases.** We have complete control over the size of our sample — the only limitation beting our time and financial constraints.
- The closer $\mathbf{X}_h$ is to the average of the sample's predictor values, the narrower the interval.

Prediction Interval for a New Response

In this section, we are concerned with the prediction interval for a new response y_{new} when the predictor values are $\mathbf{X}_h = (1, X_{h,1}, X_{h,2}, \ldots, X_{h,p-1})^{\mathrm{T}}$. Again, let's just jump right in and learn the formula for the prediction interval. The general formula in words is as always:

Sample estimate ± (*t*-multiplier × standard error)

and the formula in notation is:

$$\hat{y}_h \pm t_{(\alpha/2,n-p)} \times \sqrt{MSE + [\mathrm{se}(\hat{y}_h)]^2}$$

where:

- $\hat{y}_h$ is the "**fitted value**" or "**predicted value**" of the response when the predictor values are $\mathbf{X}_h$.
- $t_{(\alpha/2,n-2)}$ is the "***t*-multiplier**." Note again that the *t*-multiplier has *n*-*p* degrees of freedom, because the prediction interval uses the mean square error (*MSE*) whose denominator is *n*-*p*.
- $\sqrt{MSE + [\mathrm{se}(\hat{y}_h)]^2}$ is the "**standard error of the prediction**," which is very similar to the "standard error of the fit" when estimating μ_Y. The standard error of the prediction just has an extra MSE term added that the standard error of the fit does not. (More on this a bit later.)

Understanding the difference in the two formulas

Observe that the only difference in the formulas is that the standard error of the prediction for y_{new} has an extra *MSE* term in it that the standard error of the fit for μ_Y does not.

What's the practical implications of the difference in the two formulas?

- Because the prediction interval has the extra *MSE* term, a (1-α)100% confidence interval for μ_Y at $\mathbf{X}_h$ will always be narrower than the corresponding (1-α)100% prediction interval for y_{new} at $\mathbf{X}_h$.
- By calculating the interval at the sample means of the predictor values and increasing the sample size n, the confidence interval's standard error can approach 0. Because the prediction interval has the extra *MSE* term, the prediction interval's standard error cannot get close to 0.

Example

A hospital administrator wished to study the relation between patient satisfaction (Y), patient's age (X1, in years) and severity of illness (X2, an index). The administrator randomly selected 23 patients and collected the data in **patsat**, where larger values of Y and X2 are, respectively, associated with more satisfaction and increased severity of illness.

Confidence interval and prediction interval

Suppose we want to make the predictions for the follows two sets of x_0's

$$age = 50, sev = 60;$$

$$age = 45, sev = 50;$$

```
*Set up data set for predictions;
DATA c1;
sat=.; age=50; sev=60; output;
sat=.; age=45; sev=50;output;
DATA c; set c c1;
PROC PRINT data=c;
RUN;
```

Confidence interval and prediction interval

```
PROC REG data=c;
model sat=age sev /clm cli;
output out=mean_resp p=predicted stdp=SE_mean lclm =
LCL_mean uclm=UCL_mean;
output out=predict p=predicted stdi=SE_pred lcl=LCL_pred
ucl=UCL_pred;
PROC PRINT data=mean_resp; where sat=.;
PROC PRINT data=predict; where sat=.;
RUN;
```

- Run the regression model. Produce confidence limits for mean response, and prediction intervals. Put the CIs and PIs into data sets called "mean_resp" and "predict";
- "where" statements are used to select only data having a certain characteristic.

Output

Obs	sat	age	sev	predicted	SE_mean	LCL_mean	UCL_mean
24	.	50	60	38.2094	5.00948	27.7598	48.6589
25	.	45	50	55.4048	2.78061	49.6046	61.2051

Obs	sat	age	sev	predicted	LCL_pred	UCL_pred	SE_pred
24	.	50	60	38.2094	14.5822	61.8365	11.3267
25	.	45	50	55.4048	33.4346	77.3750	10.5324

Confidence interval for mean response

- Confidence interval for the mean response when age=50, sev=60:

$$\hat{y}_0 = 38.21, s\{\hat{y}_0\} = 5.01, d.f. = 20, \alpha = .05, t_{20,.025} = 2.086$$

Therefore, the 95% CI for the mean response is

$$38.21 \pm 2.086 \times 5.01 = (27.76, 48.66)$$

Prediction Interval for a New Response

- Prediction interval for the mean response when age=50, sev=60:

  $\hat{y}_0 = 38.21, s\{\hat{y}_0\} = 11.33, d.f. = 20, \alpha = .05, t_{20,.025} = 2.086$

  Therefore, the 95% CI for the mean response is

  $$38.21 \pm 2.086 \times 11.33 = (14.58, 61.84)$$